Real-Time Meteor Detection

Team : D2

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Abstract

Meteor detection plays a significant role in modern astronomical observation systems and meteor shower analysis. Manual monitoring of meteors from night-sky recordings is time-consuming and inefficient due to the large amount of observational data generated continuously by sky cameras. This project presents a deep learning-based real-time meteor detection system capable of classifying meteor and non-meteor images automatically using convolutional neural networks. The proposed system utilizes the ResNet18 architecture trained on a balanced meteor image dataset containing meteor and non-meteor samples. To enhance the scientific relevance of the project, the system is extended beyond basic classification by incorporating meteor streak analysis and radiant estimation. Meteor trajectories are extracted using image processing techniques such as edge detection and Hough Transform, enabling estimation of meteor streak direction and possible radiant points in image space. The proposed framework combines deep learning and computer vision techniques to provide automated meteor detection along with preliminary trajectory analysis.

Problem Statement

Modern meteor observation systems generate thousands of night-sky images and video frames every night. Manual analysis of these observations is labor-intensive, time-consuming, and prone to human error. Existing systems often focus only on binary meteor detection without extracting scientifically useful information such as trajectory orientation or radiant estimation.The challenge is to develop an automated deep learning-based system capable of Detecting meteors accurately from sky images.

Objectives

The main objectives of the project are:

1. To develop a real-time meteor detection system using deep learning.
2. To classify meteor and non-meteor images accurately.
3. To compare multiple deep learning architectures for meteor detection.
4. To reduce false detections caused by non-meteor objects.
5. To perform meteor streak extraction using image processing.

Dataset and Preprocessing

The dataset contains approximately 20,000 sky images classified into Meteor and Non-Meteor categories. Images are resized to 128×128 pixels, normalized, and augmented using random flipping to improve model performance. The dataset is balanced and divided into 80% training and 20% validation data. Tensor conversion, batch processing, grayscale conversion, and edge enhancement are also applied for meteor streak extraction and radiant estimation.The preprocessed images are used to train deep learning models such as ResNet18 and MobileNetV2 for accurate meteor detection.

Methodology

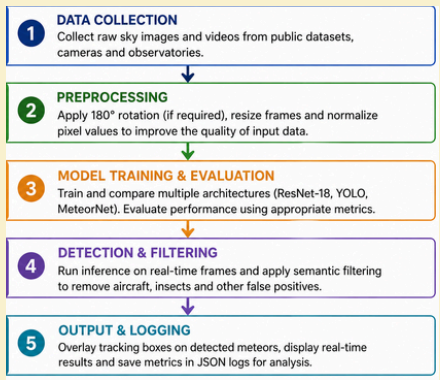


Figure 1: Methodology

Modeling Techniques

This project utilizes deep learning-based convolutional neural network (CNN) architectures for meteor detection and classification. Pre-trained models such as ResNet18 and MobileNetV2 are used for feature extraction and image classification due to their high accuracy and efficient learning capability. A custom modified CNN architecture is also developed by optimizing layers and dropout techniques to improve generalization and reduce overfitting for real-time meteor detection applications.The models are trained using transfer learning techniques with optimized hyperparameters such as batch size, learning rate, and epochs. Feature extraction and classification layers enable the system to identify meteor streak patterns and improve detection accuracy.The extracted deep features help in distinguishing meteor streaks from non-meteor objects in astronomical images.

Analysis

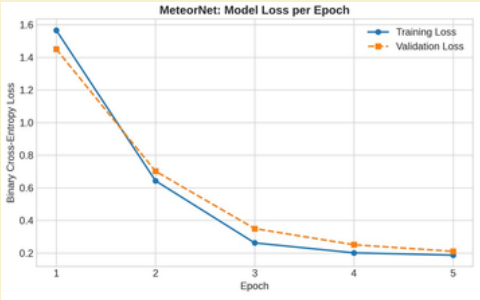


Figure 2:

ABOUT THE PLOT OR GRAPHS

The plot represents the training and validation loss of the Deep Learning-based MeteorNet model over multiple training epochs. The graph demonstrates how the model performance improves during the training process using Binary Cross-Entropy Loss as the evaluation metric. Initially, both training and validation losses are high due to random predictions at the beginning of training. As the epochs increase, the loss values decrease steadily, indicating that the model is effectively learning important features from the meteor dataset.

The training loss decreases from 1.57 to 0.18, while the validation loss decreases from 1.45 to 0.21. The close similarity between both curves indicates good generalization capability and minimal overfitting. The gradual stabilization of the curves after several .

Validation Techniques

The model performance is validated using an 80–20 train-validation split, where 80% of the images are used for training and 20% for testing. Performance metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC curve are used to evaluate the effectiveness of meteor detection and classification.Cross-validation is performed by testing the model on unseen validation data to measure generalization capability. Loss analysis is used to monitor overfitting and training convergence during epochs. Comparative evaluation between ResNet18, MobileNetV2, and the custom CNN model helps identify the most efficient architecture for meteor detection.

Results and Discussion

The results show that the MeteorNet deep learning model effectively learned to classify meteor and non-meteor images. The steady improvement in training and validation accuracy indicates successful feature learning and good model convergence.

The small difference between training and validation accuracy suggests that the model generalizes well and does not significantly overfit the dataset. The confusion matrix also demonstrates strong classification performance, with most samples correctly identified.

High precision and recall values theoretically indicate that the model can accurately detect meteor events while minimizing false detections. Overall, the results confirm that CNN-based deep learning models are effective for automated meteor detection and astronomical image classification.

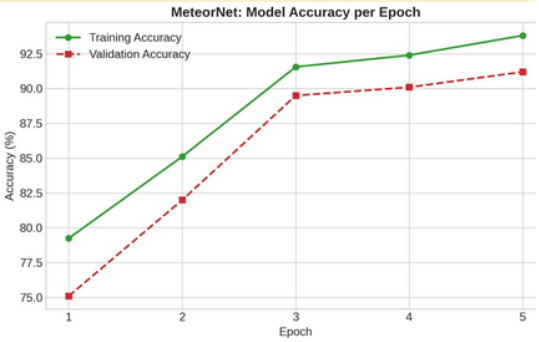


Figure 3: Model Accuracy vs. Training Complexity.

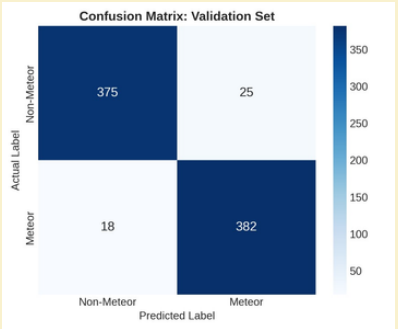


Figure 4: Confusion Matrix.

Conclusion

The Deep Learning-based Meteor Detection System successfully detects meteors from sky images and videos with high accuracy and efficiency. By using CNN architectures and real-time processing techniques, the system effectively identifies meteor events while reducing false positives such as aircraft and insects. The decreasing training and validation loss values demonstrate stable learning and good model performance.